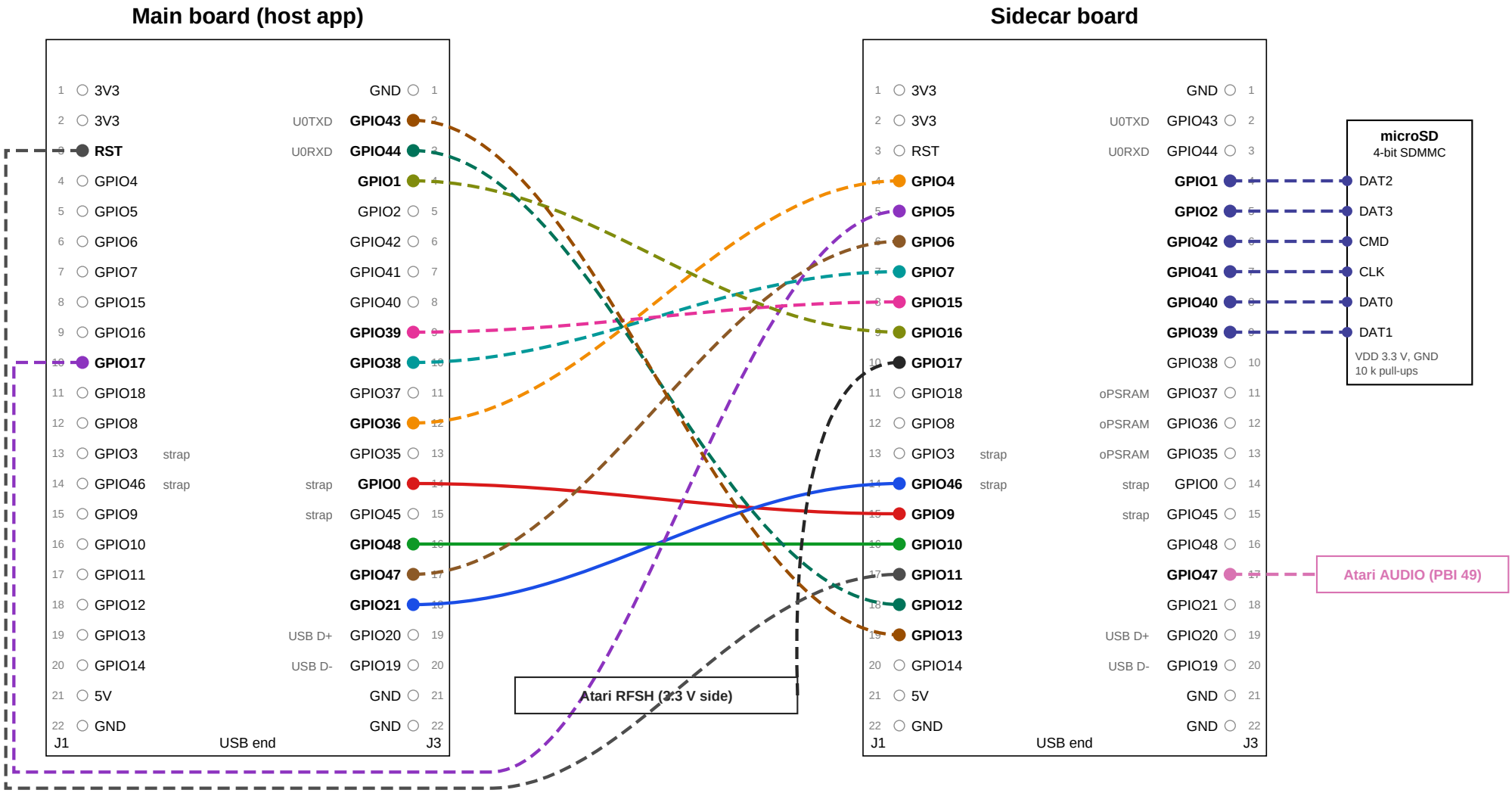
Main board <-> Sidecar wiring (ESP32-S3-DevKitC-1 pair)

Generated 2026-11-12 by hardware/tools/sidecar_wiring.py. Ground and power come through the carrier board (not drawn). Header pin numbers: DevKitC-1, pin 1 away from the USB end.



Signal	From	To	Header (from -> to)	Status	Notes
Link TX	main GPIO0	Sidecar GPIO9	J3-14 -> J1-15	wired	host TX -> Sidecar RX; also main IO0 (BOOT) for serial flashing; PBI audio on espBusBang only
Link RX	Sidecar GPIO46	main GPIO21	J1-14 -> J3-18	wired	Sidecar TX -> host RX; Sidecar carrier: via jumper J3 2-3 (1-2: GPIO21 = RFSH)
BCLK	main GPIO48	Sidecar GPIO10	J3-16 -> J1-16	wired	host drives the link clock; sampler (link off): main's profile pin GPIO48
phi2	main GPIO36	Sidecar GPIO4	J3-12 -> J1-4	planned	Sidecar bus profiler: clock
R/W	main GPIO17	Sidecar GPIO5	J1-10 -> J1-5	planned	Sidecar bus profiler; host R/W is GPIO15 on the Sidecar carrier (R/W regroup)
extSel_	main GPIO47	Sidecar GPIO6	J3-17 -> J1-6	planned	Sidecar bus profiler; open drain on the host, keep it short
D0	main GPIO38	Sidecar GPIO7	J3-10 -> J1-7	planned	Sidecar bus profiler
D1	main GPIO39	Sidecar GPIO15	J3-9 -> J1-8	planned	Sidecar bus profiler

Main board <-> Sidecar wiring (ESP32-S3-DevKitC-1 pair) (continued)

Signal	From	To	Header (from -> to)	Status	Notes
A0	main GPIO1	Sidecar GPIO16	J3-4 -> J1-9	planned	Sidecar bus profiler
RFSH	Atari RFSH	Sidecar GPIO17	- -> J1-10	planned	Sidecar bus profiler refresh_; 3.3 V side, never the 5 V bus line; or main GPIO21 where it is refresh_ (espBusBang, .5)
SD DAT2	Sidecar GPIO1	microSD DAT2	J3-4 -> -	planned	10 k pull-up
SD DAT3	Sidecar GPIO2	microSD DAT3	J3-5 -> -	planned	10 k pull-up (also CD)
SD CMD	Sidecar GPIO42	microSD CMD	J3-6 -> -	planned	10 k pull-up
SD CLK	Sidecar GPIO41	microSD CLK	J3-7 -> -	planned	up to 40 MHz; keep it short, no pull-up
SD DAT0	Sidecar GPIO40	microSD DAT0	J3-8 -> -	planned	10 k pull-up
SD DAT1	Sidecar GPIO39	microSD DAT1	J3-9 -> -	planned	10 k pull-up
Main EN	Sidecar GPIO11	main RST (EN)	J1-17 -> J1-3	planned	open drain: the Sidecar resets the main board (flashing, recovery)
Flash TX	Sidecar GPIO12	main GPIO44	J1-18 -> J3-3	planned	via 1 k to main RXD0 = Atari D6: only with the Atari off
Flash RX	main GPIO43	Sidecar GPIO13	J3-2 -> J1-19	planned	via 1 k from main TXD0 = Atari D5: only with the Atari off
Audio	Sidecar GPIO47	Atari AUDIO	J3-17 -> -	planned	via 1 k series resistor (sets the level); host GPIO0 no longer on AUDIO

Solid: wired now. Dashed: planned. Filled pin: has a wire. oPSRAM: octal-PSRAM pin (N8R8, N16R8 modules).

Bus-profiler taps: Sidecar inputs only, no pulls; 330-470 ohm in series at the main board end.
Keep the Sidecar powered (its own USB) whenever the Atari is on: an unpowered ESP32 loads the bus lines.